

Troubleshooting PC Connectivity in Router Setup

<https://gemini.google.com/app/237746a68530e813>

User prompt: My setup's like this pc0 to pc3 can communicate and transfer packets among each other but not to and from PC4 and PC5

Response: Based on the network topology and configuration screenshots, here are the potential causes for the connectivity failure to/from PC4 and PC5, along with how to fix them.

Step 1: Missing Routing Entries (Static/RIP/OSPF)

For PC0-PC3 to reach PC4 and PC5, every router must know how to reach the LAN subnet connected to **Router2** (e.g., 30.0.0.0/8 or similar) and the point-to-point link between Router1 and Router2 (50.0.0.0/8).

- **Router0** must have routes leading to Router1 and Router2's subnets.
- **Router1** must have routes leading to Router0's and Router2's subnets.
- **Router2** must have routes leading back to Router0's and Router1's subnets.

If you are using **Static Routing**, ensure the following static routes exist:

- **Router0:**

```
ip route <Router2_LAN_Network> <Subnet_Mask> 40.40.40.2
```

- **Router1:**

```
ip route <Router2_LAN_Network> <Subnet_Mask> 50.50.50.2
```

- **Router2:**

```
ip route 10.0.0.0 255.0.0.0 50.50.50.1
```

```
ip route 20.0.0.0 255.0.0.0 50.50.50.1
```

```
ip route 40.0.0.0 255.0.0.0 50.50.50.1
```

To verify success: Execute `show ip route` in the CLI tab on Router0, Router1, and Router2. Confirm that each router lists routes to all non-directly connected subnets.

Step 2: Unconfigured Interface on Router2 or PC IP Settings

Ensure the FastEthernet interface on Router2 connecting to Switch2 and the PCs (PC4 & PC5) are properly set up:

1. Open **Router2** -> **Config** -> **FastEthernet0/0** (or whichever interface connects to Switch2).
2. Ensure **Port Status** is checked **On**.

3. Confirm the IPv4 address matches the default gateway intended for PC4 and PC5.
4. On **PC4** and **PC5**, go to **Desktop -> IP Configuration**:
 - Set **Default Gateway** to Router2's FastEthernet interface IP address.
 - Ensure their IP addresses are in the same subnet as Router2's FastEthernet interface.

To verify success: Open the Command Prompt on PC4 or PC5 and run `ping <Router2_Gateway_IP>`. If it succeeds, local LAN routing is active.

Step 3: Serial Interface Connection between Router1 and Router2

Looking at the serial connection between Router1 and Router2:

1. On **Router1**, check **Serial3/0** (or whichever port connects to Router2):
 - Confirm **Port Status** is **On**.
 - Ensure the IP address is set to `50.50.50.1 / 255.0.0.0`.
2. On **Router2**, **Serial2/0** is configured as `50.50.50.2 / 255.0.0.0`.
 - Ensure **Port Status** is **On**.

To verify success: From Router1's CLI, run `ping 50.50.50.2`. A successful ping confirms the serial link between Router1 and Router2 is operational.

To help narrow down the exact issue:

1. What IP address and Subnet Mask are configured on **PC4**, **PC5**, and **Router2's LAN interface**?
2. Are you using **Static Routing** or a dynamic routing protocol (like **RIP**)?

User prompt: my faculty had told me and showed me this much only and its working fine on his machine but not in mine so show fix

Response: Since the configuration works fine on your faculty's machine, the routing logic itself (such as RIP or Static routes) is likely already set up. The issue in Cisco Packet Tracer usually comes down to **missing interface clocking, turned-off ports, or mismatched IP configurations** that were enabled on your faculty's setup but missed on yours.

Check these direct fixes step-by-step:

1 Turn On Serial Interface Port Status

1 min

On **Router1** and **Router2**, open the **Config** tab and select the Serial interfaces connecting them:

- On **Router1** (e.g., Serial3/0): Ensure **Port Status** is checked **On**.
- On **Router2** (Serial2/0): Ensure **Port Status** is checked **On**.

How to verify: Check the link between Router1 and Router2 on the topology screen. The red jagged line indicators should turn green.

2 Set Clock Rate on the DCE Serial Side

1 min

Serial link cables in Packet Tracer require one router to act as the DCE device providing clocking:

- Click on the Serial interface connected between **Router1** and **Router2** (e.g., Serial2/0 on Router2).
- Set **Clock Rate** to 2000000 (or 64000).

How to verify: Open the CLI tab on Router2 and run `show controllers serial2/0` to confirm clock rate is applied and the interface state shows "UP".

3 Verify Default Gateways on PC4 and PC5

2 min

- Click **PC4 -> Desktop -> IP Configuration**.
- Ensure **Default Gateway** is set to the exact IPv4 address of **Router2's LAN interface** (e.g., FastEthernet0/0).
- Repeat the same verification on **PC5**.

How to verify: Open **Command Prompt** on PC4 and type `ping <Router2_LAN_IP>`. You should receive a successful reply.

4 Verify Router2 FastEthernet Port Status

1 min

- Click **Router2 -> Config -> FastEthernet0/0** (or the interface connected to Switch2).
- Ensure **Port Status** is checked **On**.

How to verify: The link triangle between Router2 and Switch2 on the main diagram should turn green.

To tell you the exact missing command or configuration:

1. What IP Address and Subnet Mask are set on **PC4**, **PC5**, and **Router2's FastEthernet interface**?
2. Did your faculty use **RIP** or **Static Routing** under the Router configuration?

User prompt: Ok if u had to do the experiment tell me step by step how would u build it with the exact same steps (with gui only)

Response: Here is the exact step-by-step guide to building this entire network from scratch using only the Packet Tracer GUI.

Phase 1: Cable and Device Connections

1 Place Devices on Workspace

2 min

- From the bottom-left menu, drag **3 Router-PT** devices onto the workspace (Router0, Router1, Router2).
- Drag **3 Switch-PT** devices (Switch0, Switch1, Switch2) below each router.
- Drag **6 PC-PT** devices (PC0 & PC1 under Switch0; PC2 & PC3 under Switch1; PC4 & PC5 under Switch2).

How to verify: You should see 3 Routers, 3 Switches, and 6 PCs arranged in three distinct branches.

2 Connect PCs to Switches and Switches to Routers

3 min

- Select the **Connections** tab (lightning bolt).
- Choose **Copper Straight-Through** cable (solid black line).
- Click **PC0** -> FastEthernet0 -> click **Switch0** -> FastEthernet0/1.
- Repeat for **PC1** to **Switch0**, **PC2 & PC3** to **Switch1**, and **PC4 & PC5** to **Switch2**.
- Connect **Switch0** (FastEthernet2/1) to **Router0** (FastEthernet0/0) using a Straight-Through cable.
- Repeat for **Switch1** to **Router1** (FastEthernet0/0) and **Switch2** to **Router2** (FastEthernet0/0).

How to verify: The lines between PCs and Switches should turn green after a few seconds.

3 Connect Routers via Serial Links

2 min

- Select **Serial DCE** cable (red line with a small clock icon).
- Click **Router0** -> Serial12/0 -> click **Router1** -> Serial12/0.
- Click **Router1** -> Serial13/0 -> click **Router2** -> Serial12/0.

How to verify: Red dashed lines will connect Router0 to Router1 and Router1 to Router2.

Phase 2: Router Interface IP Configurations

1 Configure Router0 Interfaces

2 min

- Click **Router0** -> open **Config** tab.
- Select **FastEthernet0/0**:
 - Set **IPv4 Address**: 10.10.10.0 (or 10.10.10.100) with **Subnet Mask**: 255.0.0.0.
 - Check **Port Status**: **On**.
- Select **Serial2/0**:
 - Set **IPv4 Address**: 40.40.40.1 with **Subnet Mask**: 255.0.0.0.
 - Set **Clock Rate**: 2000000.
 - Check **Port Status**: **On**.

How to verify: The connection arrow pointing from Router0 down to Switch0 turns green.

2 Configure Router1 Interfaces

2 min

- Click **Router1** -> open **Config** tab.
- Select **FastEthernet0/0**:
 - Set **IPv4 Address**: 20.20.20.0 with **Subnet Mask**: 255.0.0.0.
 - Check **Port Status**: **On**.
- Select **Serial2/0**:
 - Set **IPv4 Address**: 40.40.40.2 with **Subnet Mask**: 255.0.0.0.
 - Check **Port Status**: **On**.
- Select **Serial3/0**:
 - Set **IPv4 Address**: 50.50.50.1 with **Subnet Mask**: 255.0.0.0.
 - Set **Clock Rate**: 2000000.
 - Check **Port Status**: **On**.

How to verify: The serial connection between Router0 and Router1 turns green.

3 Configure Router2 Interfaces

2 min

- Click **Router2** -> open **Config** tab.
- Select **FastEthernet0/0**:
 - Set **IPv4 Address**: 30.30.30.0 (or 30.30.30.100) with **Subnet Mask**: 255.0.0.0.
 - Check **Port Status**: **On**.
- Select **Serial2/0**:
 - Set **IPv4 Address**: 50.50.50.2 with **Subnet Mask**: 255.0.0.0.
 - Check **Port Status**: **On**.

How to verify: The serial link between Router1 and Router2 turns green.

Phase 3: PC IP Settings

1 Configure Branch 1 PCs (PC0 and PC1)

2 min

- Open **PC0** -> **Desktop** -> **IP Configuration**:
 - **IPv4 Address**: 10.10.10.1 | **Subnet Mask**: 255.0.0.0 | **Default Gateway**: 10.10.10.0
- Open **PC1** -> **Desktop** -> **IP Configuration**:
 - **IPv4 Address**: 10.10.10.2 | **Subnet Mask**: 255.0.0.0 | **Default Gateway**: 10.10.10.0

How to verify: Open Command Prompt on PC0 and type ping 10.10.10.0. Replies should return successfully.

2 Configure Branch 2 PCs (PC2 and PC3)

2 min

- Open **PC2** -> **Desktop** -> **IP Configuration**:
 - **IPv4 Address**: 20.20.20.1 | **Subnet Mask**: 255.0.0.0 | **Default Gateway**: 20.20.20.0
- Open **PC3** -> **Desktop** -> **IP Configuration**:
 - **IPv4 Address**: 20.20.20.2 | **Subnet Mask**: 255.0.0.0 | **Default Gateway**: 20.20.20.0

How to verify: Open Command Prompt on PC2 and type ping 20.20.20.0. Replies should return successfully.

3 Configure Branch 3 PCs (PC4 and PC5)

2 min

- Open **PC4** -> **Desktop** -> **IP Configuration**:
 - **IPv4 Address**: 30.30.30.1 | **Subnet Mask**: 255.0.0.0 | **Default Gateway**: 30.30.30.0
- Open **PC5** -> **Desktop** -> **IP Configuration**:
 - **IPv4 Address**: 30.30.30.2 | **Subnet Mask**: 255.0.0.0 | **Default Gateway**: 30.30.30.0

How to verify: Open Command Prompt on PC4 and type ping 30.30.30.0. Replies should return successfully.

Phase 4: Dynamic Routing Configuration (RIP)

1 Enable RIP Routing on All Routers

3 min

- Click **Router0** -> **Config** tab -> click **RIP** under **ROUTING**:
 - In the **Network** field, type 10.0.0.0 and click **Add**.
 - Type 40.0.0.0 and click **Add**.
- Click **Router1** -> **Config** tab -> click **RIP**:
 - Add 20.0.0.0.
 - Add 40.0.0.0.
 - Add 50.0.0.0.
- Click **Router2** -> **Config** tab -> click **RIP**:
 - Add 30.0.0.0.
 - Add 50.0.0.0.

How to verify: Go to **PC0** -> **Desktop** -> **Command Prompt** and type ping 30.30.30.1 (PC4). The first packet may time out during ARP, but subsequent packets will succeed.
